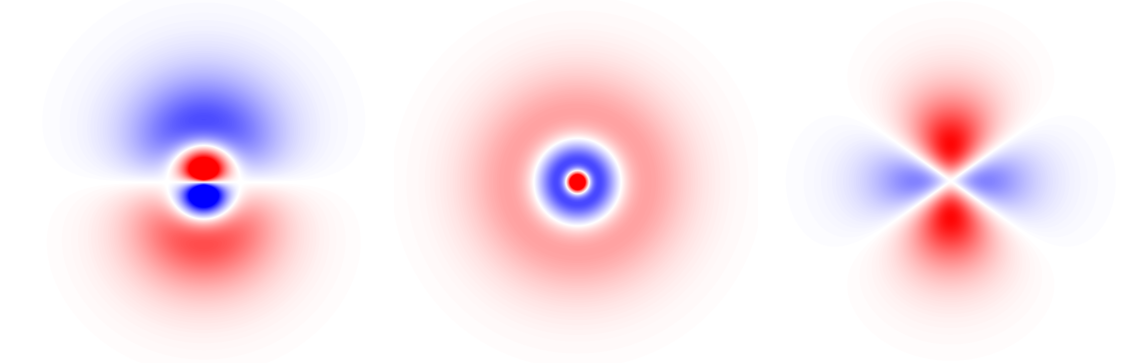


Solving the Schrödinger Equation for a Hydrogen Atom



1 Introduction

Although we often imagine subatomic particles as behaving like tiny billiard balls, the reality of the situation is rather different. In actuality, a subatomic particle is “smeared out” over multiple locations at once as described by its wavefunction ψ . If the potential energy of the particle is given by the function $U(x, y, z)$, then the wavefunction can be calculated by solving the Schrödinger equation

$$-\frac{\hbar^2}{2m} \left[\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right] + U(x, y, z)\psi = E\psi, \quad (1.1)$$

where $\hbar = h/2\pi$ is the reduced Planck’s constant, and E is the particle’s energy.

The Born interpretation of the wavefunction tells us that it is related to the probability of finding the particle in any particular location by the formula

$$|\psi(x, y, z)|^2 dx dy dz = \left(\begin{array}{l} \text{Probability of finding the particle in an} \\ \text{infinitesimal box with side lengths } dx, dy, \\ dz, \text{ centered on } (x, y, z) \end{array} \right). \quad (1.2)$$

Since the total probability of finding the particle somewhere must be equal to one, this implies that the wavefunction must be normalised, meaning that it satisfies

$$\int |\psi(x, y, z)|^2 dx dy dz = 1, \quad (1.3)$$

where the volume integral is taken over all space. Since the Schrödinger equation (1.1) is linear in the wavefunction ψ , any solution multiplied by a constant will also be a solution. Thus, so long as the integral in (1.3) is finite, we can always rescale the wavefunction so that it is appropriately normalised.

Interestingly however, most solutions of the Schrödinger equation (1.1) are non-normalisable, meaning that the wavefunction is divergent and the integral in (1.3) is infinite. Normalisable solutions are only obtained for certain very special values of E , and so we say that the energy is quantised to only take those values.

2 The Harmonic Oscillator

To demonstrate these ideas, we shall consider the behaviour of a quantum particle in a one dimensional harmonic oscillator, such as a mass on a quantum spring. In this case the potential energy is $U(x) = kx^2/2$, so the Schrödinger equation can be written as

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{kx^2}{2} \psi = E\psi. \quad (2.1)$$

To simplify matters somewhat, we shall approach the problem in a specially chosen set of units. Rather than using metres, seconds, and kilograms, we choose our units so that we have the numerical relationship $\hbar = k = m = 1$. This simplifies our Schrödinger equation into

$$-\frac{1}{2} \frac{d^2\psi}{dx^2} + \frac{x^2}{2} \psi = E\psi. \quad (2.2)$$

To solve this equation we can use a method based on educated guessing. Through a bit of trial and error we can discover that

$$-\frac{1}{2} \frac{d^2}{dx^2} \left(e^{-x^2/2} \right) + \frac{x^2}{2} e^{-x^2/2} = \frac{1}{2} e^{-x^2/2}. \quad (2.3)$$

In other words we have found a solution

$$E = \frac{1}{2} \quad \psi(x) = N e^{-x^2/2}, \quad (2.4)$$

where N is just some constant (actually it's equal to $\pi^{1/4}$) to make sure the wavefunction is normalised.

We are now going to guess that the general solution can be written in the form

$$\psi(x) = H(x) e^{-x^2/2}, \quad (2.5)$$

where $H(x)$ is some relatively simple function. At this point, we have no real reason to believe that $H(x)$ will end up being simple, and sometimes it takes a bit of trial and error to find the write guess on how to write the solution, but in this case if we substitute this expression back into the Schrödinger equation we obtain

$$\frac{d^2H}{dx^2} - 2x \frac{dH}{dx} + (2E - 1)H = 0. \quad (2.6)$$

We can solve this equation using a method known as the power series solution. We simply assume that $H(x)$ can be written in the form

$$H(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots \quad (2.7)$$

If we substitute this into equation (2.6) above and consider each power of x individually we find that

$$(k+2)(k+1)a_{k+2} - 2ka_k + (2E-1)a_k = 0, \quad (2.8)$$

which can be rearranged to yield the recursion relation

$$a_{k+2} = -\frac{2E-1-2k}{(k+2)(k+1)} a_k. \quad (2.9)$$

If we want our wavefunction to be normalisable, the power series for $H(x)$ needs to be finite, which means that there must be some integer n such that a_n is the last non-zero coefficient in the expansion. Since a_{n+2} must therefore be zero, this implies that

$$0 = \frac{2E - 1 - 2n}{(n+2)(n+1)} a_n \implies E = n + \frac{1}{2}. \quad (2.10)$$

In order to find the full solution, we note that the only way to get a finite polynomial is if the only non-zero powers are the same parity (odd or even) as n , and then just set either a_0 or a_1 to one and use the recursion relation to generate the rest of the powers. This gives us

$$\begin{aligned} H_0(x) &= 1 \\ H_1(x) &= x \\ H_2(x) &= 1 - 2x^2 \\ H_3(x) &= x - \frac{2x^3}{3} \\ H_4(x) &= 1 - 4x^2 + \frac{4x^4}{3} \\ &\vdots \end{aligned} \quad (2.11)$$

Note that the decision to start the coefficients at 1 is ultimately unimportant since a different choice would just multiply the whole function by a constant, which would get swept away in normalisation anyway.

Thus, we can write the solutions to the Schrödinger equation as:

$$\begin{aligned} E_0 = 1/2 \quad \psi_0(x) &= N_0 e^{-x^2/2} \\ E_1 = 3/2 \quad \psi_1(x) &= N_1 x e^{-x^2/2} \\ E_2 = 5/2 \quad \psi_2(x) &= N_2 (1 - 2x^2) e^{-x^2/2} \\ E_3 = 7/2 \quad \psi_3(x) &= N_3 (3x - 2x^3) e^{-x^2/2} \\ E_4 = 9/2 \quad \psi_4(x) &= N_4 (3 - 12x^2 + 4x^4) e^{-x^2/2} \\ &\vdots \end{aligned} \quad (2.12)$$

where we have absorbed the denominators into the normalisation constants N_n for notational convenience.

The only thing which remains is to transition back from our special system of units back into standard SI. To do this we need to find out what an energy of 1 in our units corresponds to in SI units. All we do is multiply the constants $\hbar$, k , and m together until we get an energy

$$1 \text{ (special units)} = \hbar \sqrt{\frac{k}{m}}, \quad (2.13)$$

which implies that in SI units the allowed energies of the harmonic oscillator are

$$E = \left(n + \frac{1}{2}\right) \hbar \sqrt{\frac{k}{m}}. \quad (2.14)$$

3 The Hydrogen Atom

When two charged particles, such as an electron and a proton, interact, their potential energy is given by the formula

$$U = \frac{Qq}{4\pi\epsilon_0 r}, \quad (3.1)$$

where Q and q are their charges, r is the distance between them, and ϵ_0 is a constant known as the permittivity of free space. This allows us to write down the Schrödinger equation for an electron in a hydrogen atom as

$$-\frac{\hbar^2}{2m_e} \left[\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right] - \frac{e^2}{4\pi\epsilon_0 \sqrt{x^2 + y^2 + z^2}} \psi = E\psi, \quad (3.2)$$

where, because the proton is so much more massive, we assume that it remains fixed at the origin. It turns out that this equation is much easier to solve if we express it in terms of spherical polar coordinates, defined by the relationships

$$x = r \sin \theta \cos \phi \quad y = r \sin \theta \sin \phi \quad z = r \cos \theta, \quad (3.3)$$

and if we also adopt the so called atomic units where numerically $\hbar = m_e = e = 4\pi\epsilon_0 = 1$, the equation takes the form

$$-\frac{1}{2} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} \right] - \frac{1}{r} \psi = E\psi. \quad (3.4)$$

Your task is to find the values of E which allow this equation to have normalisable solutions, and therefore to determine what the possible energy levels of an electron in a hydrogen atom are. You will therefore also determine the functional forms of the wavefunctions allowing you to explain the shapes of the different orbitals, as well as enabling you to answer questions such as why there is no 1p orbital.

Hints:

- It will probably be easiest to start by considering only s orbitals, which are spherically symmetric, meaning they depend only on the radius r . In this case equation (3.4) simplifies down to

$$-\frac{1}{2} \frac{d^2 \psi}{dr^2} - \frac{1}{r} \frac{d\psi}{dr} - \frac{1}{r} \psi = E\psi.$$

- The reason spherical polar coordinates make the equation easier to solve is that in these coordinates the equation becomes separable. This means that the solution can be written in the product form

$$\psi(r, \theta, \phi) = R(r)T(\theta)I(\phi),$$

allowing you to work entirely with single variable functions.

- While it is possible to stumble into appropriate guesses for the forms of the solutions, it is much easier if you have a few wavefunctions to look at so that you can spot the patterns. Hyperphysics has a nice selection of orbital wavefunctions to get you started.

<http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/hydwf.html>